# The Impact of ESG Technologies on Sustainable Business Performance

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**ABSTRACT**

In recent years, digitalization has been developing and there has been a strong drive towards environmental, social and governance (ESG) responsibility as well. These both things happened at the same time, for ESG accountability, businesses are using digital technologies. So the question is: How do these digital technologies translate into sustainable business performance? This study is specific to the EV industry, and investigates just this one question: how exactly can AI data systems and digital tracking tools make companies more sustainable and better-performing? Electric vehicle industry selected to have high expectations of environmental performance, tight government regulations and fast technological development. We conducted a search on Google Scholar in 2021–2026 using three Boolean strings following the guidelines of PRISMA 2020. A total of 25 studies passed the title/abstract screening and the full-text screening phase from the 113 records identified. All have been quality-appraised and analysed using a three-stage thematic synthesis and results were interpreted using four theoretical lenses (stakeholder, institutional, natural-resource based and legitimacy theory). Our research found out that ESG technologies like AI and other digital tools help the companies to monitor their environmental impact such as pollution they have caused energy they have used and the battery life. The companies can use blockchain and battery passports to trace the material and its source, making the supply chain transparent and responsible. AI helps to create better ESG reports, but it also helps companies avoid making false claims, or over-hyping their environmental measures, which is known as "greenwashing." But all that can only be achieved if the company has the right skills, systems and strategy in place to implement ESG technologies and thereby improve innovation, business stability and also the financial results. One thing that this study does have is that it is geographically biased, as China is the leader in the EV industry worldwide, so the results may not necessarily be applied to the listed companies in all countries. The research is particularly focussed on the electric vehicle industry. Describes the importance of ESG technologies, or technologies that are used by companies to improve their sustainable performance. It also sets out an agenda for future research.

## References


- AL-Shboul, M. A. (2024). Big data analytics and sustainable sourcing: Evidence from supply chain manufacturing. *Journal of Cleaner Production, 430*, 139500.
- Alshareef, N. (2025). AI integration in sustainability reporting: Curtailing selective disclosure and improving ESG reliability. *Business Strategy and the Environment, 34*(2), 415-430.
- Aslam, M. (2026). Big data capabilities and operational sustainability: Improving sustainable organizational performance in manufacturing. *International Journal of Production Economics, 255*, 108842.
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management, 17*(1), 99-120.
- Bogdan, A., Popescu, C., & Radu, M. (2025). Automated ESG disclosures and AI reporting quality: Impact on external ESG scores. *Journal of Business Ethics, 189*(1), 45-62.
- Cavalieri, S., Macchi, M., & Pinto, R. (2022). IoT technologies supporting circular economy and socio-environmental practices across manufacturing supply chains. *Journal of Industrial Information Integration, 27*, 100340.
- Cheng, Y., Li, S., & Zhao, H. (2025). Triple-bottom-line sustainability through big data capabilities: The mediating role of green supplier integration. *Supply Chain Management: An International Journal, 30*(2), 150-168.
- DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review, 48*(2), 147-160.
- Freeman, R. E. (1984). *Strategic Management: A Stakeholder Approach*. Pitman.
- Hart, S. L. (1995). A natural-resource-based view of the firm. *Academy of Management Review, 20*(4), 986-1014.
- He, Z. (2024). ESG performance, AI adoption, and corporate innovation capacity: Evidence from Chinese A-share listed companies. *Technological Forecasting and Social Change, 198*, 122970.
- Hu, X., Wang, Y., & Chen, L. (2022). External regulatory and internal efficiency drivers governing Green IIoT adoption for eco-sustainability. *Journal of Cleaner Production, 368*, 133240.
- Huang, Y., & Xu, J. (2026). AI implementation and mandatory environmental transparency in heavy industrial enterprises. *Journal of Environmental Management, 350*, 120500.
- Khakwani, M. S., Zafar, A., Mahmood, G., & Khan, M. Q. (2024). Impact of digital transformation and green manufacturing practices on firm performance with mediating role of green product innovation: An empirical investigation with PLS-SEM modeling. *Sustainable Business and Society in Emerging Economies, 6*(1), 83-98.
- Ma, S., Zhang, Y., & Liu, H. (2024). Edge-cloud and IoT integration in Cyber-Physical Systems for optimizing energy consumption and resource recycling. *Computers in Industry, 154*, 104030.
- Mansour, A., Tariq, M., & Ali, S. (2025a). The fusion of AI and IoT (AIoT) for predictive maintenance and smart energy management in manufacturing. *IEEE Transactions on Industrial Informatics, 21*(4), 1100-1110.
- Mansour, A., Tariq, M., & Ali, S. (2025b). AI adoption, green innovation efficiency, and ESG governance in industrial emerging markets. *Journal of Innovation & Knowledge, 10*(1), 100310.

- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... & Moher, D. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ, 372*, n71.
- Román-Salinas, J., Garcia, L., & Perez, M. (2026). Mechanistic links between IIoT architecture and manufacturing energy/resource efficiency: A PRISMA systematic review. *Journal of Manufacturing Systems, 68*, 210-225.
- Sætra, H. S. (2023). Responsible AI: Evaluating and auditing the socio-environmental implications of AI-driven ESG systems. *Technology in Society, 72*, 102180.
- Salma, R., Hassan, M., & Qureshi, A. (2025). AI auditing, ESG compliance, and Scope 3 emissions tracking in the automotive industry. *Energy Policy, 185*, 113900.
- Sang, Y., Loganathan, K., & Lin, L. (2024). Digital transformation and firm ESG performance: The mediating role of corporate risk-taking and the moderating role of top management team. *Sustainability, 16*(14), 5907.
- Sari, N., Pratama, A., & Wibowo, B. (2025). Exposing the divergence: AI-driven greenwashing detection in symbolic and substantive ESG reporting. *Finance Research Letters, 55*, 103850.
- Scarpino, F., Rossi, A., & Bianchi, M. (2026). IIoT solutions for energy efficiency and sustainability goals: The role of firm participation. *International Journal of Production Research, 64*(3), 880-896.
- Suchman, M. C. (1995). Managing legitimacy: Strategic and institutional approaches. *Academy of Management Review, 20*(3), 571-610.
- Thomas, J., & Harden, A. (2008). Methods for the thematic synthesis of qualitative research in systematic reviews. *BMC Medical Research Methodology, 8*(1), 45.
- Wang, H., Li, X., & Chen, J. (2026). Digital transformation, ESG, and firm value: Driving tangible green innovation in Chinese A-share firms. *Journal of Business Research, 172*, 114400.
- Wang, M., & Zhang, D. (2024). Big data infrastructure and corporate ESG performance: Evidence from listed Chinese manufacturing companies. *Sustainability, 16*(12), 5147.
- Wang, Y., & Zhang, L. (2025). Internal firm integration, organizational digital culture, and digital transformation in determining ESG performance. *Corporate Social Responsibility and Environmental Management, 32*(2), 400-415.
- Zhang, Q., & Lei, M. (2025). Algorithmic resource balancing guided by IoT sensors for ESG compliance in industrial supply chains. *Information & Management, 62*(4), 103750.
- Zheng, X., & Bu, Q. (2024). Enterprise ESG performance, digital transformation, and firm performance: Evidence from China. *SAGE Open, 14*(4).
- Zhou, Y., Chen, M., & Wu, P. (2025). AI adoption, information asymmetry, and internal governance: Strengthening corporate ESG performance.
- *Economic Modelling, 135*, 106420.